Smart Fuel ka poora flow

Is page par project ka har hissa aasaan alfaaz mein hai: kaun kya karta hai, har button dabane ke baad andar kya hota hai, har API route kya karta hai aur code use kaise sambhalta hai, kaun se rules lagte hain, data kahan save hota hai, aur viva mein kya poocha ja sakta hai.

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SRS ki saari functional requirements ka code mukammal (Phase 0 se Phase 6). Ab sirf testing aur submission (Phase 7).

FR-01	Account: register, login, profile	FR-02	Vehicle add karna	FR-03	Vehicle dekhna, edit, delete
FR-04	Fuel purchase records	FR-05	Trips banana	FR-06	Fuel ka andaza aur behtar raasta
FR-07	Crisis mode	FR-08	Monthly budget aur alerts	FR-09	Charts aur graphs
FR-10	Dashboard filters	FR-11	Map par raasta	FR-12	Map zoom, pan, click
FR-13	Notifications				

IS PAGE PAR

- Ek nazar mein
- System ka dhancha
- Ek request ka safar
- Code kahan hai
- Har API route
- Kaun kya kar sakta hai
- Har feature ka flow
  - Account
  - Vehicles
  - Fuel records
  - Monthly budget
  - Reports
  - Trips
  - Map
  - Crisis mode
  - Admin
  - Notifications
- Database
- Saare hisaab
- Hifazat
- Testing
- Project chalana
- Viva ki tayyari
- Kya baqi hai

Ek nazar mein

Pakistan mein fuel mehnga hai aur crisis ke dino mein kam bhi ho jata hai. Smart Fuel ek web app hai jis mein log aur chhoti companies apni gaariyon ka fuel, kharcha aur safar ek jagah sambhalte hain, aur app unhein fuel bachane ke mashware deti hai.

Individual

Aam user. Apni ek ya zyada gaariyan, apni fuel entries, apna budget, apne trips.

Organizational

Company ka account. Ek login se poori fleet (kai gaariyan). Fleet ka budget aur chahein to har gaari ka alag budget.

Admin

System chalane wala. Saare users aur gaariyan dekhta, theek aur delete karta hai. Register page se nahi banta.

APP KE PAGES

- **Dashboard**: account ki maloomat aur is mahine ka fuel budget.
- **Vehicles**: gaariyan add, edit, delete.
- **Fuel Records**: har dafa fuel bharwane ki entry.
- **Trips**: safar plan karna, raaste map par.
- **Crisis mode**: fuel bachane ke mashware.
- **Ghanti (top bar)**: notifications: budget, mehenga raasta, crisis mashware.
- **Reports**: charts aur gaariyon ka muqabla.
- **Profile**: apni maloomat aur password.
- **Administration**: sirf Admin ke liye.

KIS CHEEZ SE BANA HAI

- **Frontend**: React 19 aur Vite 7 (browser mein chalne wala hissa).
- **Backend**: Node.js, Express 5 (server jo faisle karta hai).
- **Database**: MongoDB Atlas, Mongoose 8 ke zariye.
- **Map**: Leaflet aur OpenStreetMap ki tasveeren (tiles).
- **Raaste aur jagah**: OpenRouteService (free API).
- **Charts**: Recharts.
- **Toasts**: react-toastify.

MERN ka matlab: MongoDB, Express, React, Node. Chaaron JavaScript mein hain, is liye rules ek hi zubaan mein dono taraf chal jate hain.

System ka dhancha

Project ke teen bade hisse hain. Browser sirf dikhata aur poochta hai; har asal faisla server karta hai; data database mein rehta hai.

BROWSER

React app

Pages, forms aur map. Form ke rules foran check karta hai taa ke user ko wahin pata chale. Login ka token apne paas rakhta hai.

`frontend/`



SERVER

Express API

Har request par check: kaun hai, kya ijazat hai, data sahi hai. Phir kaam karta hai aur JSON jawab deta hai.

`backend/ · /api/...`



DATABASE

MongoDB

Users, gaariyan, fuel entries, budgets, trips aur notifications. Sirf server is se baat karta hai.

`6 collections`

Browser aur Server dono

shared/ rules

Saari limits aur check ek jagah (`shared/constants.js` , `shared/validation.js`), aur rupay, date aur jagah likhne ka tareeqa (`shared/format.js`). Form aur server dono yahi use karte hain, is liye dono ka jawab aur likhai hamesha ek jaisi.

Sirf Server

OpenRouteService

Jagah dhoondna aur raaste nikalna. Is ki API key sirf server ki `.env` mein; browser tak kabhi nahi jati.

Neeche har flow mein har qadam par likha hai ke woh kahan hota hai: AAP BROWSER SERVER DATABASE BAHAR

Sirf Browser

(OpenRouteService)

Map ki tasveeren (tiles). Browser seedha OSM se leta hai; is ke liye key nahi chahiye.

Ek request ka safar

Misaal: aap fuel entry save karte hain. Har feature isi tarah chalta hai; sirf beech ke rules badalte hain.

1. AAP — **Form bhar kar Save dabate hain.**
2. BROWSER — **Form ke rules check hote hain.**
   Ghalti ho to field ke neeche laal message; request jati hi nahi.
3. BROWSER — **`api.js` token laga kar request bhejta hai.**
   `POST /api/fuel-records`, header mein `Authorization: Bearer <token>`.
4. SERVER — **Darwaze ke checks.**
   `helmet` (security headers), `cors` (sirf apna frontend), JSON parhna.
5. SERVER — **`protect`: token sahi hai?**
   Token check, phir user ko database se dobara parhna. Is liye admin role badle to foran lagu hota hai.
6. SERVER — **Controller asal kaam karta hai.**
   Rules dobara (browser par bharosa nahi), malikana haq (apni gaari?), aur khaas rules (tank, odometer, qeemat).
7. DATABASE — **Mongoose record save karta hai.**
8. SERVER — **Notification check.**
   Is entry se us mahine ka budget 80% ya 100% tak pohncha to ghanti ke liye ek notification (pehle se ho to nahi).
9. SERVER — **Jawab ek hi shakal mein.**
   Kamyabi: `{ success: true, message, record }`. Ghalti: `errorHandler` se `{ success: false, message }`.
10. BROWSER — **Screen update hoti hai.**
    List aur budget alerts dobara load, aur ghanti foran nayi notifications check karti hai. Jawab 401 ho (token khatam) to app khud sign out kar deti hai.

CODE	MATLAB	KAB AATA HAI
400	Ghalat input	Rule toota, jaise khali field ya ghalat date
401	Pehchan nahi	Token nahi ya expire; app sign out kar deti hai
403	Ijazat nahi	Doosre ki gaari, ya admin budget par
404	Nahi mila	Record delete ho chuka
409	Pehle se maujood	Wahi email ya registration number
422	Raasta nahi	Do jagahon ke beech sadak nahi mili
429	Bohat koshishein	Login, register ya route service ki had
503	Route service band	Key nahi lagi, ya OpenRouteService jawab nahi de rahi
500	Server ki ghalti	Andar ki tafseel user ko nahi dikhai jati, sirf server log mein

Code kahan hai

`backend/` aur `frontend/` do alag npm projects hain. Root ka `npm run dev` dono ek saath chalata hai.

JAGAH	KYA KAAM
`backend/server.js`	Server shuru: settings check, database connect, middleware, saare routes
`backend/config/`	`env.js` zaroori settings check karta hai (JWT_SECRET 32+ characters); `db.js` MongoDB se jodta hai
`backend/routes/`	Kaun sa URL kis controller par jaye, aur kaun sa middleware lage
`backend/middleware/`	`auth.js` (protect, adminOnly, nonAdminOnly), `rateLimit.js`, `errorHandler.js`
`backend/controllers/`	Asal kaam: auth, user, vehicle, fuelRecord, budget, report, trip, place, crisis, notification, admin
`backend/models/`	Database ki shakal: User, Vehicle, FuelRecord, Budget, Trip, Notification

JAGAH	KYA KAAM
backend/utils/	Chhote helpers: access.js (malikana haq), fuelReport.js (mileage, report), tripEstimate.js (fuel ka andaza), ors.js (route service), crisisAdvice.js (crisis mashware), notifications.js (notification banana), budgetStatus.js (budget ki haalat aur mahine ka kharcha), ttlCache.js , token.js , dates.js , numbers.js
backend/seeders/	adminSeeder.js : pehla Admin banana
backend/dev/fakeOrs.js	Nakli route service: tests aur bina internet demo ke liye
frontend/src/App.jsx	Pages ke raaste; login hai to app, warna login/register
frontend/src/api.js	Server se baat: token lagana, 401 par sign out, har save ke baad ghanti ko batana
frontend/src/components/	Har page, form aur card (Vehicles.jsx, TripForm.jsx, CrisisMode.jsx, NotificationBell.jsx...)
frontend/src/hooks/	Data lana aur form sambhalna: useAuth , useForm , useFetch , useTrips , useCrisisAdvice , useNotifications ...
frontend/src/constant.js validation.js · format.js	Labels aur menu; shared rules ka darwaza; rupay, litre, date ki formatting
frontend/src/toast.js	Toasts (**react-toastify** library): delete se pehle upar chhota sawal (ConfirmToast.jsx : Delete/Cancel, Escape = Cancel) aur kaam ke baad 4 second ka success message. Browser ka confirm() box kahin nahi; ToastContainer Layout mein
frontend/src/global.css	Saari styling, light aur dark theme
shared/	constants.js aur validation.js : dono taraf ke rules; format.js : rupay, date aur jagah ki likhai (server notification ke message mein bhi)
docs/	SRS, roadmap, architecture report

Har API route

Browser server se in 43 raaston (routes) ke zariye baat karta hai. Har route ke saath likha hai: woh kya karta hai, code use qadam ba qadam kaise sambhalta hai, kaun se middleware pehle chalte hain, kaun sa function kaam karta hai, aur app ka kaun sa page use bulata hai.

`/health` `/auth` `/users` `/vehicles` `/fuel-records` `/budgets` `/reports` `/trips` `/places` `/crisis` `/notifications` `/admin`

EK ROUTE KA RAASTA CODE MEIN

1. `server.js` : `app.use("/api/trips", tripRoutes)` batata hai ke `/api/trips` wali har request `routes/tripRoutes.js` mein jaye.
2. Route file: method aur raasta (jaise `POST /:id/routes`), pehle middleware, phir controller ka function.
3. Controller: asal kaam, aur aakhir mein JSON jawab.
4. Koi ghalti "throw" ho to Express 5 khud `errorHandler` tak le jata hai, is liye controllers mein `try/catch` nahi.
5. Koi route match na ho: 404 "Route not found".

HAR CONTROLLER KA EK HI DHANCHA

1. **Whitelist**: `pick...Fields(req.body)` sirf jaane pehchane fields leta hai; baqi gira deta hai (koi `user` ya `role` chupke se nahi bhej sakta).
2. **Rules**: `firstError(check..., check...)` ; pehli ghalti par 400.
3. **Dhoondna**: `findById` ; na mile to 404.
4. **Malikana haq**: `isOwnerOrAdmin(req, record.user)` ; na ho to 403. Lists mein `ownerFilter(req)` : user ko apne, admin ko sab.
5. **Khaas rules**: tank, odometer, budget ka jor, waghaira.
6. **Save aur jawab**: `{ success: true, message, ... }` .

MIDDLEWARE	KYA KARTA HAI	KAHAN LAGTA HAI
`protect`	Header mein `Bearer` token dekhta hai, `jwt.verify` se check, phir user database se (password ke baghair) `req.user` mein. Token nahi: 401 "Authentication required". Ghalat ya expire: 401. User delete ho chuka: 401 "User no longer exists". Database ki ghalti 500 deti hai, 401 nahi, taa ke user sign out na ho.	Login, register aur health ke ilawa har route
`adminOnly`	Role Admin nahi to 403.	`/api/admin`
`nonAdminOnly(message)`	Admin ho to 403 aur diya hua message.	`/api/budgets` , `/api/crisis` , `/api/notifications` , trips badalna (`viewOnlyForAdmins`)

MIDDLEWARE	KYA KARTA HAI	KAHAN LAGTA HAI
`loginLimiter`	Ek IP se 15 minute mein 10 nakaam login; phir 429. Kamyab login nahi gine jate.	`POST /auth/login`
`registerLimiter`	Ek IP se 1 ghante mein 10 registrations.	`POST /auth/register`
`placeLimiter`	Har user 10 minute mein 30 jagah search.	`/api/places`
`routeLimiter`	Har user 10 minute mein 20 raaste wali requests.	Trip banana, edit, recalculate
`errorHandler`	Har ghalti ko `{ success: false, message }` banata hai: ghalat JSON 400, ghalat id 400 "Invalid id", schema ki ghalti 400, unique ka takrao 409, baqi 500 (tafseel sirf server log mein).	Sab se aakhir mein, har route ke liye

/api/health

`server.js`

GET **/api/health**

Server chal raha hai ya nahi.

1. Seedha `{ success: true, message: "Smart Fuel API is running" }`. Login nahi chahiye.

Kahan se: testing aur check ke liye

/api/auth

`routes/authRoutes.js → controllers/authController.js`

Login se pehle wale routes; token yahin banta hai.

POST **/api/auth/register**

Naya Individual ya Organizational account.

1. Rules: naam, email, phone, role (sirf Individual ya Organizational; "Admin" rad), Organizational ho to company ka naam, password (8–64, letters aur numbers). Ghalti: 400.
2. Email chhote huroof mein; pehle se maujood: 409 "Email already registered".
3. `bcrypt.hash(password, 12)`, phir `User.create`.
4. 201: `token` (`createToken`, 1 din) aur `user.toPublic()`.

Middleware: registerLimiter **Function:** register **Kahan se:** AuthPage.jsx (Register)

POST /api/auth/login

Email aur password se andar aana.

1. Email aur password text hon aur khali na hon; warna 400.
2. Email se user dhoondna, `bcrypt.compare` se password milana.
3. User na mile ya password ghalat: 401 "Invalid email or password" (dono haalat mein ek hi message, taa ke koi andaza na laga sake ke email maujood hai).
4. Kamyab: `token` aur `user.toPublic()`.

Middleware: loginLimiter **Function:** login **Kahan se:** AuthPage.jsx (Login)

POST /api/auth/logout

Sign out.

1. Server sirf "Logged out successfully" kehta hai. Asal sign out browser karta hai: `useAuth` token aur user mita deta hai, server ke jawab ka intezar kiye baghair.

Middleware: protect **Function:** logout **Kahan se:** hooks/useAuth.js (Sign out)

/api/users

`routes/userRoutes.js → controllers/userController.js`

GET /api/users/profile

Apni taaza maloomat.

1. `protect` pehle hi user load kar chuka; jawab `req.user.toPublic()` (crisisMode samet).

Middleware: protect **Function:** getProfile **Kahan se:** useAuth (app khulne aur tab par wapas aane par)

PUT /api/users/profile

Profile aur password badalna.

1. Sirf bheje gaye fields check; company ka naam sirf Organizational ke liye.
2. Naya password ho to `currentPassword` zaroori; `bcrypt.compare` se milana. Ghalat: 400 (401 nahi, warna sign out).
3. Email badla aur kisi aur ka hai: 409.
4. Trim karke save; naya password hash ho kar. Jawab: `user.toPublic()`.

Middleware: protect **Function:** updateProfile **Kahan se:** Profile.jsx

/api/vehicles

`routes/vehicleRoutes.js → controllers/vehicleController.js`

Sab par `protect`. Admin aur malik dono yahi routes use karte hain.

POST /api/vehicles

Nayi gaari.

1. `pickVehicleFields` : sirf 9 fields.
2. `vehicleTypeError` pehle (type ki range: engine, km/l, tank), phir `vehicleError` (har field). Ghalti: 400.
3. Electric ho to engine cc hata diya jata hai.
4. `normalizeVehicleText` : `lea 1234` → `LEA-1234`, `HONDA` → `Honda`.
5. `registrationTaken` : regex jo space aur hyphen ignore kare; mil jaye to 409.
6. `Vehicle.create` malik `req.user._id` ke saath; 201.

Function: createVehicle **Kahan se:** VehicleForm.jsx

GET /api/vehicles

Gaariyon ki list.

1. `Vehicle.find(ownerFilter(req))` : user ko apni, admin ko sab; har gaari ke saath malik ka naam aur email.

Function: getVehicles **Kahan se:** hooks/useVehicles.js (Vehicles, Fuel Records, Trips, Reports)

GET /api/vehicles/:id

Ek gaari.

1. `findById` ; na mile 404; malik ya admin nahi to 403.

Function: getVehicle **Kahan se:** app abhi istemal nahi karti; API mukammal rakhne ke liye

PUT /api/vehicles/:id

Gaari edit.

1. 404 aur 403 ke checks.
2. Type ki range poori gaari par (saved + naye fields mila kar), is liye purani gaari ko is edit par type deni hogi. Baqi rules sirf bheje gaye fields par.
3. Electric ho to engine hata; likhai ek jaisi.
4. Registration badla to takrao check, apni gaari chhod kar; 409.
5. `Object.assign` aur `save`.

Function: updateVehicle **Kahan se:** VehicleForm.jsx (Vehicles aur Administration)

DELETE /api/vehicles/:id

Gaari aur us ka saara data delete.

1. 404 aur 403 ke checks.
2. Us gaari ki fuel entries aur budgets delete.
3. Us ke trips ki ids nikal kar trips delete; `Trip.releaseCombinedTrips(ids)` doosri gaariyon ke combined trips ko dobara Planned karta hai.
4. Phir gaari delete.

Function: deleteVehicle **Kahan se:** Vehicles.jsx, hooks/useAdmin.js

/api/fuel-records

routes/fuelRecordRoutes.js → controllers/fuelRecordController.js

Sab par `protect`.

POST /api/fuel-records

Nayi fuel entry.

1. `pickFuelFields`; "odometer not working" na bheja ho to `false`.
2. `checkRecord`: date (aaj tak), litre aur rupay 0 se zyada, reading ya "not working", station ki lambai, aur price per litre Rs 100–1,000. Ghalti: 400.
3. `findFuelVehicle`: gaari ki id sahi (400), gaari maujood (404), apni (403), electric nahi (400).
4. `checkTankCapacity`: litre tank se zyada: 400.
5. Reading ho to `checkOdometerOrder`: pichli tareekhon ki sab se bari reading aur agli tareekhon ki sab se chhoti reading se compare; message mein us entry ki date.
6. Naya record, malik = gaari ka malik. `applyFields`: date UTC midnight par; "not working" ho to reading khali.
7. `save`: model ka `pre("validate")` hook price per litre = rupay ÷ litre khud likhta hai.
8. `notifyBudgetLimits(record.user, mahina)`: us mahine ka koi budget 80% ya 100% tak pohncha to gaari ke malik ko notification (har budget, har mahina ek dafa). 201.

Function: createFuelRecord **Kahan se:** FuelRecordForm.jsx

GET /api/fuel-records?vehicle=&from=&to=

Entries ki list, filter ke saath.

1. `ownerFilter(req)`, aur chahein to gaari.
2. `from` / `to` sahi dates hon aur from ≤ to; warna 400.
3. Gaari aur malik ke saath, nayi date pehle.

Function: getFuelRecords **Kahan se:** FuelRecords.jsx

GET /api/fuel-records/:id

Ek entry.

1. 404 aur 403 ke checks; gaari aur malik ke saath.

Function: getFuelRecord **Kahan se:** app abhi istemal nahi karti

PUT /api/fuel-records/:id

Entry edit.

1. 404 aur 403 ke checks. Reading bheji aur checkbox nahi: odometer chal raha samjha jata hai.
2. Saved values par naye values mila kar poora record `checkRecord` se.
3. Gaari ya litre badle tabhi gaari ke rules (electric, tank); gaari badli to entry ka malik bhi badalta hai.
4. Reading, date ya gaari badle tabhi odometer ki tarteeb check, is entry ko chhod kar.
5. `applyFields` aur `save` (price dobara nikalti hai).
6. `notifyBudgetLimits` dobara, kyunke edit se kharcha barh sakta hai.

Function: updateFuelRecord **Kahan se:** FuelRecordForm.jsx

DELETE /api/fuel-records/:id

Entry delete.

1. 404 aur 403 ke checks, phir `deleteOne`. Budget aur reports khud theek, kyunke woh har dafa jod kar bante hain.

Function: deleteFuelRecord **Kahan se:** FuelRecords.jsx

/api/budgets

`routes/budgetRoutes.js → controllers/budgetController.js`

Sab par `protect` aur `nonAdminOnly` ("Admin accounts do not have budgets"). Har query sirf `req.user` ke budgets par.

GET /api/budgets?month=2026-10

Ek mahine ka budget aur kharcha.

1. Mahina na diya to is mahine (Pakistan ka waqt); ghalat shakal: 400.
2. Ek saath teen kaam: us mahine ke budgets, har gaari ka kharcha (`loadMonthSpending` : MongoDB `aggregate` se jod), aur Organizational ho to us ki gaariyan.
3. `compareSpendingWithBudget` : baqi rakam, % aur haalat (`getBudgetStatus` : ok, warning 80%+, over 100%+).
4. `editable` : mahina badla ja sakta hai ya nahi.

Function: getMonthlyBudgetSummary

Kahan se: hooks/useMonthlyBudgetSummary.js (BudgetCard, BudgetAlerts)

PUT /api/budgets

Budget set ya edit (ek hi route).

1. `checkBudgetMonth` (is mahine se 12 aage tak) aur `checkBudgetAmount` (0 se zyada, 1 crore tak). Ghalti: 400.
2. Gaari nahi bheji (fleet budget): naya amount gaariyon ke budgets ke jor se kam: 400.
3. Gaari bheji: sirf Organizational; gaari apni aur electric nahi; fleet budget pehle se ho; sab gaariyon ka jor (is gaari ka purana chhod kar, naya jod kar) fleet se zyada: 400.
4. `findOneAndUpdate(..., { upsert: true })`: hai to badlo, nahi to banao.
5. `notifyBudgetLimits`: kam kiya hua budget foran 80% ya 100% par ho sakta hai.

Function: createOrUpdateBudget **Kahan se:** BudgetForm.jsx

DELETE /api/budgets/:id

Budget hatana.

1. 404; kisi aur ka: 403; pichla mahina: 400.
2. Fleet budget hai aur us mahine gaariyon ke budgets maujood: 400 "Remove the vehicle budgets for this month first".
3. `deleteOne`.

Function: deleteBudget **Kahan se:** BudgetCard.jsx

/api/reports

`routes/reportRoutes.js → controllers/reportController.js + utils/fuelReport.js`

GET /api/reports?from=&to=&vehicle=&fuelType=&user=

Charts aur gaariyon ke muqable ka saara data.

1. `readReportFilters`: `to` na ho to is mahina; `from` na ho to 12 mahine pehle; range check (120 mahine tak); gaari, fuel type ki jaanch; `user` sirf admin ka maana jata hai. Ghalti: 400.
2. Gaariyan: `ownerFilter(req, userId)` + gaari + fuel type.
3. Un gaariyon ki range wali entries, purani pehle (mileage ke liye zaroori).
4. `chooseGrouping`: 24 mahine se lamba ho to saal ke hisaab se.
5. `sumRecords` (kul), `sumRecordsByPeriod` (khali mahine 0 ke saath), `buildVehicleStats` (har gaari, `calculateMileage` se).

Middleware: protect **Function:** getFuelReport **Kahan se:** hooks/useFuelReport.js (Reports)

/api/trips

routes/tripRoutes.js → controllers/tripController.js + utils/ors.js, utils/tripEstimate.js

Sab par `protect`. Badalne wale routes par `viewOnlyForAdmins` (admin 403); raaste mangwane walon par `routeLimiter`. Har single trip ka jawab `withDetails` se: gaari, malik, aur combine ke links.

GET /api/trips?vehicle=&status=&user=

Trips ki list.

1. `user` sirf admin ka; ids sahi hon; status chaaron mein se ho (planned, completed, postponed, combined). Ghalti: 400.
2. `select("-routes.geometry")` : list mein raaston ki lambi lines nahi, taa ke jawab halka rahe.
3. Gaari aur malik ke saath, nayi date pehle.

Function: getTrips **Kahan se:** hooks/useTrips.js (Trips)

GET /api/trips/estimate-basis?vehicle=

Form ke liye gaari ki km/l aur aakhri qeemat.

1. `findTripVehicle` : gaari zaroori, sahi id, maujood, apni.
2. `loadEstimateBasis` : pichle 12 mahine ki entries → `buildEstimateBasis` : 300 km se zyada naape gaye to odometer wali mileage, warna likhi; tank range; aakhri entry ki qeemat aur date.

Function: getEstimateBasis **Kahan se:** hooks/useEstimateBasis.js (TripForm qeemat khud bharta hai)

POST /api/trips

Naya trip, raaston aur fuel ke andaze ke saath.

1. `pickTripFields` ; `findTripVehicle` .
2. `checkTrip` : dono jagah Pakistan mein, aapas mein 0.2 km se door, date aaj se 365 din tak, priority, purpose 200 tak, qeemat (electric ke ilawa). Ghalti: 400.
3. Naya `Trip` : malik = gaari ka malik, status hamesha planned, electric ki qeemat null.
4. `calculateRoutes` : andaze ki buniyad save; `findRoutes` (ORS se 3 raaste ek saath); `addFuelEstimates` (litre, rupay, refuel); `chooseRecommended` ; chuna hua = recommended. Service fail ho to `routeError` likh kar aage; kabhi rukta nahi.
5. `save` ; `notifyAboutTrip` (crisis mode on ho aur yeh trip kisi combine ya postpone mashware mein aaye to notification); 201. Message mein "routes could not be calculated: ..." agar raaste nahi mile.

Middleware: viewOnlyForAdmins, routeLimiter **Function:** createTrip **Kahan se:** TripForm.jsx

GET /api/trips/:id

Ek trip, map ki lines ke saath.

1. `findById` ; 404; malik ya admin nahi to 403; `withDetails` (gaari, malik, `combinedInto` , `combinedTrips`).

Function: getTrip **Kahan se:** hooks/useTrip.js (TripDetails)

PUT /api/trips/:id

Trip edit, raasta chunna, ya Mark completed.

1. 404 aur 403. Trip postponed ya combined ho: 400 "plan it again before changing it".
2. Gaari badli to `findTripVehicle` . Saved + naye values par `checkTrip` ; date ka rule sirf tab jab date ya status badle (purana trip edit ho sake).
3. `selectedRoute` bheja to trip ke raaston mein se ho; warna 400.
4. Gaari ya jagah badli: `calculateRoutes` (naye raaste). Sirf qeemat badli: `addFuelEstimates` se naye rupay, ORS ko call nahi.
5. `save` . Pehli dafa completed hua to us mein combine kiye trips bhi completed.
6. `notifyAboutTrip` : chuna hua raasta recommended se 15% ya zyada fuel le to "inefficient trip" notification (har trip ek dafa); crisis mode on ho to naye mashware ki bhi.

Middleware: viewOnlyForAdmins, routeLimiter **Function:** updateTrip

Kahan se: TripForm.jsx (Edit), TripDetails.jsx (raasta, Mark completed)

DELETE /api/trips/:id

Trip delete.

1. 404 aur 403; `deleteOne` ; `Trip.releaseCombinedTrips([id])` us mein combine kiye trips ko dobara Planned karta hai.

Middleware: viewOnlyForAdmins **Function:** deleteTrip **Kahan se:** Trips.jsx, TripDetails.jsx

POST /api/trips/:id/routes

Raaste dobara mangwana (Recalculate, Try again).

1. 404 aur 403; sirf planned trip (completed ka andaza wahi rehta hai), warna 400; gaari dhoond kar `calculateRoutes` ; `save` ; `notifyAboutTrip` .

Middleware: viewOnlyForAdmins, routeLimiter **Function:** recalculateTripRoutes **Kahan se:** TripDetails.jsx

POST /api/trips/:id/plan-again

Crisis mode ka faisla wapas (Postponed ya Combined → Planned).

1. 404 aur 403; sirf postponed ya combined trip, warna 400.
2. `ignoredSuggestions` mein "postpone" ya "combine" jod diya, taa ke wahi mashwara dobara na aaye.
3. Status planned, `combinedInto` hata; `save` ; `notifyAboutTrip` . Date guzar chuki ho to message: date badlein.

Middleware: viewOnlyForAdmins **Function:** planTripAgain **Kahan se:** TripDetails.jsx (Plan again)

/api/places

`routes/placeRoutes.js → controllers/placeController.js + utils/ors.js`

Sab par `protect` aur `placeLimiter` . OpenRouteService ko sirf `ors.js` bulata hai; ghalti kabhi 401 nahi banti (503, 429...).

GET /api/places/search?text=

Jagah ka naam likh kar dhoondna.

1. `checkPlaceSearch` : 3 se 100 characters; warna 400.
2. `findPlaces` : pehle cache (7 din); nahi to ORS se sirf Pakistan ke 5 nateeje; naam se ", Pakistan" hata diya.
3. Service ki ghalti ho to us ka status aur saaf message.

Function: searchPlaces **Kahan se:** hooks/usePlaceSearch.js (PlaceField)

GET /api/places/reverse?lat=&lng=

Map par click ki jagah ka naam.

1. lat aur lng numbers hon aur Pakistan ke andar; warna 400 "Pick a point in Pakistan".
2. `findPlaceAt` : qareeb koi naam ho to woh, warna `null` (form "Pinned point ..." rakhta hai).

Function: reversePlace **Kahan se:** TripForm.jsx (map click ke baad)

/api/crisis

routes/crisisRoutes.js → controllers/crisisController.js + utils/crisisAdvice.js

Sab par `protect` aur `nonAdminOnly` ("Admin accounts do not have crisis mode"). Mashware kabhi save nahi hote; har request par trips se naye bante hain.

GET /api/crisis

Mashware, bachat aur budget ki jaanch.

1. Mode off: `{ crisisMode: false, advice: null }`.
2. Ek saath: `Trip.findUpcomingPlanned` (aaj aur aage ke planned trips) aur `loadBudgetLeft` (is mahine ka fleet budget − `loadMonthSpending` ka kharcha).
3. `buildCrisisSuggestions`: combine, phir postpone, phir raasta; har ek ki bachat.
4. Is mahine ke trips: `compareTripsWithBudget`; `totalSaving`; ignore kiye mashware hain ya nahi.

Function: getCrisisAdvice **Kahan se:** hooks/useCrisisAdvice.js (CrisisMode)

PUT /api/crisis/mode

Crisis mode on ya off.

1. `enabled` true ya false ho, warna 400; `User.findByIdAndUpdate`; jawab mein naya `user` (browser use save karke topbar ki nishani dikhata hai). On karne par notification nahi banti; page khud mashware dikhata hai.

Function: setCrisisMode **Kahan se:** CrisisMode.jsx (switch)

POST /api/crisis/suggestions/:id/accept

Mashwara maan lena.

1. `findSuggestion`: mode off ho to 400; mashware abhi ke trips se dobara bana kar yeh id dhoondi; na mile (trips badal gaye) to 404.
2. combine: baqi trips `combined`, `combinedInto` = main trip. postpone: `postponed`. route: `selectedRoute` = recommended.

Function: acceptSuggestion **Kahan se:** CrisisMode.jsx (Accept)

POST /api/crisis/suggestions/:id/ignore

Mashwara chhupana.

1. `findSuggestion` (upar jaisa); mashware ke har trip ke `ignoredSuggestions` mein us ki qisam (`$addToSet`, dobara nahi judti). Trips nahi badalte.

Function: ignoreSuggestion **Kahan se:** CrisisMode.jsx (Ignore)

DELETE /api/crisis/ignored

Ignore kiye mashware wapas.

1. User ke `ignoredSuggestions` hata diya (`$unset`).

Function: showIgnoredSuggestions **Kahan se:** CrisisMode.jsx ("Show ignored suggestions again")

/api/notifications

`routes/notificationRoutes.js → controllers/notificationController.js · banti hain utils/notifications.js mein`

Sab par `protect` aur `nonAdminOnly` ("Admin accounts do not have notifications"). Har query sirf `req.user` ki notifications par. Notification is API se nahi banti; upar wale routes save ke baad khud banate hain.

GET /api/notifications

Ghanti ki list aur unread ginti.

1. Ek saath do kaam: nayi 50 notifications (`type, message, link, read, createdAt`; andar ki `key` nahi bheji jati) aur saari unread ki ginti (`countDocuments`).

Function: getNotifications

Kahan se: hooks/useNotifications.js (har minute, window par wapas aane par, aur har save ke baad)

PUT /api/notifications/:id/read

Ek notification parh li.

1. `findOneAndUpdate({ _id, user: req.user._id }, { read: true })`: id ke saath user bhi milaya, is liye kisi aur ki notification par 404 (yeh bhi pata nahi chalta ke woh maujood hai). Ghalat id: 400.

Function: markNotificationRead **Kahan se:** NotificationBell.jsx (notification par click)

PUT /api/notifications/read-all

Sab parh li.

1. `updateMany({ user, read: false }, { read: true })`.

Function: markAllNotificationsRead **Kahan se:** NotificationBell.jsx ("Mark all as read")

/api/admin

`routes/adminRoutes.js → controllers/adminController.js`

Sab par `protect` aur `adminOnly`. Admin gaariyan edit aur delete aam `/api/vehicles/:id` se karta hai; yahan sirf list aur search.

GET /api/admin/users

Saare users.

1. Password ke baghair, naye pehle. Search browser mein hoti hai.

Function: getUsers **Kahan se:** hooks/useAdmin.js, Reports.jsx (admin ka user filter)

GET /api/admin/users/:id

Ek user.

1. Password ke baghair; na mile 404.

Function: getUser **Kahan se:** hooks/useAdmin.js

PUT /api/admin/users/:id

User edit, role samet.

1. Bheje gaye fields ke rules; naya role Organizational ho to company ka naam zaroori. Ghalti: 400.
2. Admin apna Admin role hataye: 400. Email kisi aur ka: 409.
3. Role set; Organizational nahi to company ka naam saaf; Admin bane to crisis mode off; `save`.
4. Saaf: Admin bane to saare budgets aur notifications; Individual bane to gaariyon ke budgets.

Function: updateUser **Kahan se:** UserForm.jsx

DELETE /api/admin/users/:id

User aur us ka saara data delete.

1. 404; apna account: 400; phir us ki fuel entries, budgets, trips, notifications, gaariyan, aur aakhir mein user.

Function: deleteUser **Kahan se:** hooks/useAdmin.js

GET `/api/admin/vehicles?user=&registrationNumber=`

Saari gaariyon mein search.

1. Registration: `registrationNumberPattern` (space aur hyphen ignore; "LEA 12" bhi "LEA-1234" dhoond leta hai).
2. User: naam ya email mein (chhote bade huroof barabar; text `escapeRegex` se mehfooz), un users ki ids.
3. Gaariyan malik ke saath, nayi pehle.

Kaun kya kar sakta hai

Ijazat ka faisla server ke controller mein hota hai. Browser sirf button chhupata hai; asal rok server par hai.

Function: searchVehicles **Kahan se:** hooks/useAdmin.js (typing rukne ke 250 ms baad)

KAAM	INDIVIDUAL	ORGANIZATIONAL	ADMIN
Register	Haan	Haan (org ka naam zaroori)	Nahi; seeder ya doosra admin
Vehicles	Apni	Apni fleet	Sab ki
Fuel records	Apne	Fleet ke	Sab ke
Monthly budget	Ek budget	Fleet + har gaari	Nahi
Reports	Apna data	Fleet ka muqabla	Sab ka ya ek user ka
Trips	Apne	Fleet ke	Sirf dekhna
Crisis mode	Apne liye	Apne liye	Nahi
Notifications	Apni	Apni (poori fleet ki)	Nahi
Users sambhalna	Nahi	Nahi	Haan

> Admin ka trips "sirf dekhna" roadmap ki proposal hai. Supervisor se confirm karna baqi hai.

Har feature ka flow

Har feature mein pehle qadam ba qadam flow, phir us ke rules, phir khaas haalat (edge cases), aur aakhir mein woh files jahan code hai.

Account: register, login, profile FR-01

1. AAP — **Register form bharte hain.**
   Naam, email, phone, account type (Individual ya Organizational), Organizational ho to company ka naam, aur password.
2. BROWSER — **Form ke rules.**
   Password 8 se 64 characters, letters aur numbers dono.
3. SERVER — **Rules dobara.**
   "Admin" account type rad. Wahi email pehle se ho to 409.
4. SERVER — **Password ka hash.**
   bcrypt (12 rounds). Asal password kahin save nahi hota.
5. DATABASE — **User save.**
6. SERVER — **Token banta hai.**
   JWT, 1 din ke liye, saath mein user ki public maloomat (password nahi).
7. BROWSER — **Token aur user localStorage mein; Dashboard khulta hai.**
   Login bhi isi tarah: email aur password, phir token. Logout par browser token mita deta hai.

RULES

- 15 minute mein 10 ghalat login (ek IP se) ke baad 429. Register: 1 ghante mein 10.
- Har request par server user ko database se dobara parhta hai.
- App khulne aur tab par wapas aane par profile dobara load hoti hai, taa ke admin ki tabdeeli dikhe.
- Profile: naam, email, phone, company ka naam badal sakte hain.
- Naya password sirf purana password de kar.

KHAAS HAALAT

- Purana password ghalat: 400, 401 nahi; warna app user ko sign out kar deti.
- Kisi bhi request par 401 (token expire, user delete): app khud sign out.
- Admin register se nahi banta: `npm run seed:admin` (`.env` ki ADMIN_* values) ya Administration page.
- Seeder dobara chalane se maujooda admin ka password nahi badalta.
- Purane 6-character password wale users bhi login kar sakte hain; naya password naye rule par.

Files: authController.js, userController.js, middleware/auth.js, utils/token.js · hooks/useAuth.js, AuthPage.jsx, Profile.jsx

Vehicles FR-02 FR-03

1. AAP — **Vehicles → Add vehicle.**
   Naam, registration, type, manufacturer, model year, fuel type, engine cc, fuel efficiency (km/l), tank (litre).
2. BROWSER — **Type ke hisaab se range check.**
   Field chhodte hi us ka error dikhta hai (khali field ka error sirf Save par).
3. SERVER — **Pehle type ki range, phir har field.**
4. SERVER — **Ek jaisi likhai.**
   `lea 1234` → `LEA-1234`, `HONDA` → `Honda`.
5. SERVER — **Registration kisi aur ke paas to nahi?**
   Space aur hyphen chhod kar compare; mil jaye to 409.
6. DATABASE — **Save; list mein naya card.**

RULES

- 6 types: Motorcycle, Rickshaw, Car, SUV/Jeep, Van/Pickup, Bus/Truck. Har type ki engine, km/l aur tank ki range. Misaal Car: 600–3,000 cc, 6–30 km/l, 25–80 litre.
- Electric gaari: engine cc nahi; baqi range lagu.
- Registration: letters se shuru, zyada se zyada 4 hisse, har hissa 4 characters tak (LEA-1234, LEB-07-1234, ICT-AB-123).
- Model year 1980 se agle saal tak.
- km/l engine ke hisaab se ajeeb lage to sirf peeli warning; save phir bhi hota hai.

KHAAS HAALAT

- Number box mouse wheel se nahi badalta; `1e3` jaisa text rad.
- Purani gaariyan jin ki type nahi: list mein "Type missing"; agli edit par type deni hogi.
- Delete: us ki fuel entries, budgets aur trips bhi delete. Kisi aur gaari ke trips jo is ke trip mein combine the, dobara Planned.
- Doosre user ki gaari: 403.

Files: vehicleController.js, models/Vehicle.js · Vehicles.jsx, VehicleForm.jsx, hooks/useVehicles.js

Fuel records FR-04

1. AAP — **Fuel Records → Add entry.**
   Gaari, date, litre, kul rupay, odometer reading (ya "My odometer is not working"), station.
2. BROWSER — **Rules check.**
3. SERVER — **Date ka rule.**
   `YYYY-MM-DD`, aaj tak; "aaj" Pakistan ke waqt se.
4. SERVER — **Gaari ke rules.**
   Litre tank se zyada nahi. Electric gaari ki fuel entry nahi.
5. SERVER — **Odometer peeche nahi ja sakta.**
   Pichli tareekhon ki sab se bari reading se kam nahi; agli tareekhon ki sab se chhoti se zyada nahi.
6. DATABASE — **Save, qeemat khud nikalti hai.**
   Price per litre = kul rupay ÷ litre. Rs 100 se 1,000 ke beech hona chahiye (warna typing ki ghalti).
7. BROWSER — **List aur budget alerts dobara load.**

RULES

- Price per litre user nahi bhejta; server khud nikalta hai.
- Entry ka malik gaari ka malik (admin add kare tab bhi).
- Date us din ki UTC midnight par save hoti hai, is liye har jagah din wahi rehta hai.
- Gaari ke hisaab se filter.

KHAAS HAALAT

- Ek hi din ki do entries (top-up) aapas mein compare nahi hotin.
- Edit par tank aur electric rule sirf tab jab gaari ya litre badle; odometer rule jab reading, date ya gaari badle. Is tarah purani entries edit ho sakti hain.
- Reading ke baghair entry ka matlab "odometer not working".

Files: fuelRecordController.js, models/FuelRecord.js · FuelRecords.jsx, FuelRecordForm.jsx

Monthly budget aur alerts FR-08

1. AAP — **Dashboard ke "Fuel budget" card mein mahina chunein → Set budget.**
2. SERVER — **Mahina aur rakam check.**
   Is mahine se 12 mahine aage tak; rakam 0 se zyada, Rs 1 crore tak.
3. DATABASE — **Budget save.**
   User + mahina + gaari. Gaari khali = poore mahine (fleet) ka budget.
4. SERVER — **Har dafa dekhne par kharcha jodta hai.**
   Us mahine ki fuel entries ka kul.
5. SERVER — **Haalat.**
   Kharcha ÷ budget × 100: 80% se "warning", 100% se "over".
6. BROWSER — **Progress bar aur alert.**
   Peela (80%) ya laal (100%) alert Dashboard aur Fuel Records page par.

RULES

- Individual: har mahine ek budget.
- Organizational: fleet budget, aur chahein to har petrol/diesel gaari ka.
- Pehle fleet budget; gaariyon ke budgets ka jor fleet budget se zyada nahi.
- Budget agle mahine khud nahi chalta; har mahine naya.
- Kharcha kabhi save nahi hota, is liye entry badle to budget foran sahi.

KHAAS HAALAT

- Pichle mahine sirf dekhne ke liye.
- Gaariyon ke budgets hon to fleet budget un ke jor se kam ya remove nahi ho sakta.
- Admin ka budget nahi (403).
- Role badle: Admin bane to saare budgets delete; Individual bane to gaariyon ke budgets delete.
- Alert tab tak dikhta hai jab tak kharcha zyada hai. Is ke ilawa 80% aur 100% par ghanti mein ek dafa notification (har budget, har mahina).

Files: budgetController.js, utils/budgetStatus.js, models/Budget.js · BudgetCard.jsx, BudgetForm.jsx, BudgetAlerts.jsx, hooks/useMonthlyBudgetSummary.js

Reports, charts aur filters FR-09 FR-10

1. AAP — **Reports page par filters chunein.**
   Period (This month, 3, 6 ya 12 mahine, ya custom), gaari, fuel type. Admin user bhi chun sakta hai.
2. BROWSER — **Filters URL mein.**
   `?from=2026-05&to=2026-10&vehicle=...` ; refresh ya link share par bhi wahi report.
3. SERVER — **Matching gaariyan aur un ki entries.**
4. SERVER — **Hisaab.**
   Kul litre, kul kharcha, har mahine ka (24 mahine se lamba ho to har saal ka), har gaari ka hisaab aur mileage.
5. BROWSER — **Charts aur table.**
   Mahana litre aur kharcha; gaariyon ka muqabla (litre, kharcha, mileage). Gaari ke bar par click se us gaari ka filter.

MILEAGE KAISE NAAPI JATI HAI

- Full-tank method: do odometer readings ka farq ÷ un ke beech bhare litre.
- Kisi gap ki mileage likhi efficiency se 2 guna zyada: beech ki entries darj nahi hui, woh gap chhod diya.
- Koi istemal ke laaiq gap nahi: gaari ki likhi efficiency, grey bar "(estimate)".

KHAAS HAALAT

- Khali mahine 0 ke saath, chart mein bar nahi.
- From, To ke baad ho to error.
- Custom range 120 mahine tak.
- Charts ki library sirf Reports page khulne par download hoti hai (app tez).

Files: reportController.js, utils/fuelReport.js · Reports.jsx, ReportFilters.jsx, FuelTrendChart.jsx, VehicleComparison.jsx, hooks/useReportFilters.js, useFuelReport.js

Trips aur fuel ka andaza FR-05 FR-06

1. AAP — **Trips → Plan a trip → gaari chunein.**
2. AAP — **Start aur destination.**
   Naam likh kar Search (ya Enter) aur list se chunein, ya "Map" daba kar map par click.
3. BAHAR — **OpenRouteService jagah dhoondti hai.**
   Sirf Pakistan, 5 nateeje. Server jawab 7 din yaad rakhta hai.
4. AAP — **Date, priority, maqsad, qeemat.**
   Date aaj se 365 din aage tak. Priority: Essential ya "Can wait". Fuel ki qeemat gaari ki aakhri entry se khud bhari (badal sakte hain).
5. SERVER — **Andaze ki buniyad.**
   Pichle 12 mahine mein 300 km se zyada naape gaye to odometer wali mileage, warna gaari ki likhi km/l. Tank range = km/l × tank.
6. BAHAR — **3 raaste ek saath.**
   Fastest, Shortest, Avoids motorways. Ek jaise raaste hata diye jate hain.
7. SERVER — **Har raaste ka fuel.**
   Litre = km ÷ km/l; rupay = litre × qeemat. Tank range se lamba ho to "Refuel on the way".
8. SERVER — **Recommended chunna.**
   Sab se kam litre wala raasta, bashart woh sab se tez raaste se 1.5 guna se zyada waqt na le.
9. DATABASE — **Trip andaze ke saath save.**
   Baad mein qeemat ya gaari badle to purana andaza wahi rehta hai, jab tak Recalculate na karein.
10. BROWSER — **Map aur table.**
    Doosra raasta table ke radio ya line par click se chun sakte hain.

RULES

- Motorcycle aur rickshaw Pakistan mein motorway par mana hain, is liye un ke dono raaste motorway se bachte hain. Bus/truck heavy vehicle profile par.
- Electric: raaste, km aur waqt; litre aur rupay nahi.

KHAAS HAALAT

- Route service band, slow (10 second), limit khatam ya key nahi: trip phir bhi save, wajah aur "Try again".
- Route service ki ghalti kabhi 401 nahi, warna user sign out ho jata.
- Har user: 30 search aur 20 raaste fi 10 minute, kyunke free quota sab mein bata hai (2,000

- Naya trip hamesha Planned. Phir Edit, Recalculate, Mark completed, Delete.
- Edit: nayi gaari ya jagah ho to naye raaste; sirf qeemat badle to wahi raaste, naye rupay.
- Planned trip ki date guzar jaye to "Overdue".
- Chuna hua raasta recommended se 15% ya zyada fuel le: ghanti mein "Route needs more fuel".
- Khula hua trip address mein (`/trips?trip=<id>`), is liye notification seedha usi trip ko kholti hai.

raaste aur 1,000 search roz).

- OpenRouteService 100 km se lambe safar ke alternative raaste nahi deti, is liye 3 alag requests.
- Start aur destination 0.2 km se kam faasle par: rad.

Files: tripController.js, placeController.js, utils/ors.js, utils/tripEstimate.js, models/Trip.js · Trips.jsx, TripForm.jsx, PlaceField.jsx, TripDetails.jsx, RouteList.jsx

Map FR-11 FR-12

1. BROWSER — **Leaflet map khulta hai.**
   OpenStreetMap ki tiles seedha browser mein.
2. BROWSER — **Pins aur lines.**
   Start aur destination ke gol pins; har raaste ki line (server se aaya encoded text, `polyline.js` kholta hai).
3. BROWSER — **Chuna hua raasta upar.**
   Moti amber line; baqi grey.
4. AAP — **Click, pan, zoom.**
   Line par click se raasta chunna; drag se map hilana; +/− ya keyboard se zoom.

RULES

- Mouse wheel se zoom sirf map par click ke baad, taa ke page scroll karte map na hile.
- Map Pakistan ke gird; zoom 5 se kam nahi.
- Map par OpenStreetMap aur openrouteservice ka naam (attribution) hamesha.
- Dark theme mein tiles bhi dark.

KHAAS HAALAT

- Tiles na khulein: message, aur neeche table mein saari tafseel.
- Phone par map form se upar, taa ke jagah chunne ke liye scroll na karna pare.
- Map sidebar aur popups ke upar nahi aata (CSS isolation).
- Leaflet sirf Trips page khulne par download hota hai.

Files: TripMap.jsx, polyline.js, global.css (TRIPS hissa)

Crisis mode FR-07

SRS: crisis mode "users" ko sab se zaroori kaam par dhyan dene deta hai aur safar milane ya ghair-zaroori safar chhorne jaise mashware deta hai. Mashware sirf mashware hain; faisla user ka.

1. AAP — **Crisis mode page → "Turn on crisis mode".**
2. SERVER — **User par `crisisMode` on.**
   Har page ke topbar par laal "Crisis mode on" nishani.
3. SERVER — **Aaj aur aage ke Planned trips parhta hai.**
   Mashware har dafa naye bante hain; save nahi hote.
4. SERVER — **Combine.**
   Ek hi din ke trips jin ke start 5 km ke andar aur manzilen 5 km ke andar: ek hi safar mein karein.
5. SERVER — **Postpone.**
   "Can wait" trip: fuel milna aasaan hone tak taal dein.
6. SERVER — **Raasta.**
   Chuna hua raasta recommended se lamba: recommended lein.
7. SERVER — **Bachat aur budget.**
   Har mashware ki bachat (km, litre, rupay) aur kul bachat. Is mahine ke trips ka kharcha bache budget se zyada ho to warning.
8. AAP — **Accept ya Ignore.**
9. DATABASE — **Accept ka asar.**
   Combine: baqi trips "Combined". Postpone: trip "Postponed". Raasta: recommended chun liya. Ignore: trip par yaad rakha jata hai. Kuch delete nahi hota.
10. AAP — **Undo: Trips page par "Plan again".**
    Trip wapas Planned.

RULES

- Har user apne liye on/off; Admin ka crisis mode nahi.
- Ek trip ek hi mashware mein: pehle combine, phir postpone, phir raasta.
- Combine ka main trip: Essential pehle, phir sab se lamba. Baqi us par chhote stop.

KHAAS HAALAT

- Koi planned trip nahi: sirf aam tips.
- Sirf ek trip: combine ka mashwara nahi.
- Electric trip ko petrol trip mein milana fuel zyada khaye: combine nahi.
- Main trip complete: combined trips bhi complete. Main trip ya us ki gaari delete: combined trips dobara Planned.

- Raaste ke baghair trip ka faasla: seedha faasla × 1.3.
- 1 km se kam bachat wala mashwara nahi dikhaya jata.
- Mode band karne par mashware chhup jate hain; trips nahi badalte.
- Mode on ho aur naya ya badla trip kisi combine ya postpone mashware mein aaye: ghanti mein notification; click par wohi mashwara highlight.
- Postponed ya Combined trip edit nahi hota jab tak "Plan again" na karein.
- "Plan again" ke baad wahi mashwara us trip par dobara nahi; "Show ignored suggestions again" sab wapas laata hai.

Files: crisisController.js, utils/crisisAdvice.js · CrisisMode.jsx, hooks/useCrisisAdvice.js, Layout.jsx (topbar), TripDetails.jsx (Plan again)

Admin SRS se zyada

1. AAP — **Admin login.**
   Account seeder ya doosre admin ne banaya.
2. BROWSER — **Sidebar mein "Administration".**
   "Crisis mode" ka link nahi.
3. AAP — **Users.**
   List (search browser mein), edit (role bhi), delete.
4. SERVER — **Gaariyon ki search.**
   User ke naam/email ya registration se; typing rukne ke 250 ms baad. Registration mein space aur hyphen ignore.
5. AAP — **Reports aur trips.**
   Sab ka ya ek user ka report; sab ke trips dekhna.

RULES

- Admin khud ko delete ya apna Admin role khatam nahi kar sakta (server aur browser dono).
- User delete: us ki gaariyan, entries, budgets, trips aur notifications bhi.
- Admin ki ghanti nahi (us ke budget, trips aur crisis mode nahi).

ROLE BADALNE PAR

- Organizational na rahe to company ka naam saaf.
- Jo budgets naya role nahi rakh sakta, delete.
- Admin bane to crisis mode off aur notifications delete.

Files: adminController.js · Admin.jsx, UserForm.jsx, hooks/useAdmin.js

Notifications FR-13

SRS: system users ko zaroori waqiyon par bataye: budget cross, inefficient trip, aur crisis mode mein mashware. App ke andar ghanti aur list; email nahi.

1. AAP — **Koi kaam save karte hain.**
   Fuel entry, budget, trip, ya trip ka raasta chunna.
2. SERVER — **Kaam save, phir notification check.**
   Controller save ke baad `notifyBudgetLimits` ya `notifyAboutTrip` bulata hai.
3. SERVER — **Waqiye ki pehchan (key).**
   Misaal: `budget-over:2026-10:overall`, `inefficient-trip:<trip id>`. Isi key se pata chalta hai ke yeh pehle bataya ja chuka hai ya nahi.
4. DATABASE — **Sirf nayi ho to save.**
   `updateOne(..., { $setOnInsert }, { upsert: true })`; `{ user, key }` ka unique index ek saath aayi do requests mein bhi doosri nahi banne deta.
5. BROWSER — **Ghanti foran check karti hai.**
   `api.js` har save ke baad `useNotifications` ko batata hai; is ke ilawa har minute aur window par wapas aane par.
6. AAP — **Laal ginti → list → click.**
   Notification "read" ho jati hai aur us cheez ka page khulta hai: budget → Dashboard, trip → wohi trip, crisis → wohi mashwara (highlight).

NOTIFICATION	KAB BANTI HAI	KITNI DAFA
Budget almost used	Fuel entry add/edit ya budget save ke baad kisi budget ka kharcha 80% ya zyada	Har budget, har mahina ek dafa
Budget exceeded	Kharcha 100% ya zyada (seedha 100% se upar jaye to sirf yahi)	Har budget, har mahina ek dafa
Route needs more fuel	Planned trip ka chuna hua raasta recommended se 15% ya zyada fuel le	Har trip ek dafa
Crisis mode suggestion	Crisis mode on, aur naya ya badla trip kisi combine ya postpone mashware mein	Har mashwara ek dafa

RULES

- Har user sirf apni notifications dekhta hai; Admin ki koi nahi.

KHAAS HAALAT

- Trip delete ho chuka: click par "Trip not found" aur Close button.

- Gaari ke malik ko notification, chahe entry admin ne daali ho.
- 60 din baad MongoDB khud mita deta hai (TTL index).
- List mein nayi 50; ginti saari unread ki.
- Notification banane mein ghalti ho to sirf server log; user ka kaam phir bhi save (500 nahi).
- Mashwara khatam (accept, ignore ya trips badle): crisis page par "no longer available".
- Crisis mode on karne par notification nahi; page khud mashware dikhata hai.
- Raaste wala crisis mashwara alag notification nahi deta; "Route needs more fuel" kaafi hai.
- Kharcha kam ho kar dobara barhe: usi mahine dobara notification nahi.

Files: models/Notification.js, utils/notifications.js, utils/budgetStatus.js, notificationController.js · NotificationBell.jsx, hooks/useNotifications.js, api.js

Database

MongoDB mein 6 collections. Ek user ki kai gaariyan; ek gaari ki kai fuel entries, budgets aur trips. Delete hamesha neeche tak jata hai.

COLLECTION	KYA RAKHTA HAI	KHAAS BAAT
`users`	naam, email, phone, role, company ka naam, password ka hash, crisisMode	Email unique
`vehicles`	user, naam, registration, type, engine cc, manufacturer, model year, fuel type, km/l, tank	Registration poore system mein unique
`fuelrecords`	user, vehicle, date, litre, kul rupay, price per litre, odometer, station	Price server khud nikalta hai
`budgets`	user, month (`"2026-10"`), vehicle (khali = fleet), amount	User + mahina + gaari unique
`trips`	user, vehicle, start, destination, date, purpose, priority, status, andaze ki buniyad (km/l, qeemat, tank range), routes, recommended/selected raasta, routeError, combinedInto, ignoredSuggestions	Raaste ki line chhote encoded text mein
`notifications`	user, type, key (waqiye ki pehchan), message, link, read, createdAt	User + key unique; 60 din baad khud delete (TTL)

Jo save nahi hota, har dafa banta hai: budget ka kharcha, report ke totals aur mileage, aur crisis ke mashware. Faida: data ek hi jagah rehta hai aur kabhi purana nahi hota. Entry badli to agle hi request par sab sahi.

Saare hisaab ek jagah

Viva mein sab se zyada yahi poocha jata hai. Har jawab andaza hai; traffic, chadhaai aur AC shamil nahi.

KYA	FORMULA	MISAAL
Price per litre	kul rupay ÷ litre	Rs 9,000 ÷ 30 L = Rs 300/L
Mileage	(agli reading − pichli reading) ÷ beech ke litre	450 km ÷ 30 L = 15 km/l
Trip ka fuel	km ÷ km/l	300 km ÷ 15 = 20 L
Trip ka kharcha	litre × qeemat	20 L × Rs 300 = Rs 6,000
Tank range	km/l × tank	15 × 40 L = 600 km
Recommended raasta	sab se kam km, jin ka waqt ≤ 1.5 × sab se tez	Fastest 4 h ho to 6 h tak ke raaste
Budget %	kharcha ÷ budget × 100	80% warning, 100% over (dono par ek dafa notification)
Inefficient trip	(chune raaste ka km ÷ recommended ka km − 1) × 100 ≥ 15	130 ÷ 105 = 24% zyada → notification
Combine ki bachat	doosre trip ka km − (dono manzilon ka seedha faasla × 1.3)	8.2 km − 1.9 km = 6.3 km
Postpone ki bachat	poore trip ka fuel	289 km ÷ 14 = 20.6 L
Raaste ki bachat	chune raaste ka km − recommended ka km	130 − 101 = 29 km
Raaste ke baghair trip	seedha faasla × 1.3	Sadak seedhi line se lambi hoti hai

"Kam fuel" ka matlab yahan "kam km" hai, kyunke litre = km ÷ km/l. Is liye Lahore se Islamabad par GT Road (chhota) Recommended aa sakta hai, motorway nahi. 1.5 guna waqt ki had is liye hai taa ke bohat slow raasta na chuna jaye.

Hifazat

PEHCHAN AUR IJAZAT

- Password bcrypt hash (12 rounds); kabhi wapas nahi bheja jata.
- JWT token 1 din; har request par user database se dobara.
- `JWT_SECRET` 32 characters se chhota ho to server start hi nahi hota.
- Malikana haq har controller mein: user sirf apna data. Notifications ki har query bhi user ke saath; kisi aur ki par 404.
- Admin register page se nahi ban sakta.

SERVER KI HIFAZAT

- `helmet` security headers; `cors` sirf apne frontend ko.
- Rate limits: login, register, jagah search aur raaste.
- Rules dono taraf; server browser par bharosa nahi karta.
- Email, registration aur notification ki key database mein bhi unique (ek saath do requests par bhi).
- Route service ki key sirf server ki `.env` mein.
- 500 ki andar ki tafseel user ko nahi dikhai jati.

Testing kaise hui

Har phase ke baad automated scripts se. Asli database ya chalte servers ko kabhi haath nahi lagaya gaya.

TAREEQA

- Memory mein chalne wala MongoDB (test ke baad khatam).
- Test API port 5100, test website port 5174.
- Nakli route service port 5199 (asli key istemal nahi hoti).
- Browser tests: Playwright aur Chrome; map tiles ki jagah khaali tasveer.
- `npm run build` se frontend ka check.

NATEEJE (13 OCT 2026)

- 399 API checks pass: notifications 57, crisis mode 65, trips 89, purane 188 (vehicles, budget, reports, fixes).
- Browser: notifications 27, crisis mode 30, trips 39, aur 5 purane pages ke tests pass.
- Console mein koi ghair-mutawaqqa error nahi.

Project chalana

Node.js 20.6 ya naya chahiye.

1 **Packages install** (pehli dafa), project ke root se:

```
npm run install:all
```

2 `backend/.env` **banayein** (`.env.example` se copy). Zaroori: `MONGODB_URI` , `JWT_SECRET` (32+ characters). Optional: `ORS_API_KEY` (raaste aur search ke liye), `ADMIN_*` (seeder ke liye).

3 **Pehla admin banayein:**

```
npm run seed:admin
```

4 **Dono servers chalayein:**

```
npm run dev
```

API: `http://localhost:5000` · Website: `http://localhost:5173`

5 **Bina internet ya key ke demo:** `backend/` mein `node dev/fakeOrs.js` , aur API ko `ORS_BASE_URL=http://localhost:5199 ORS_API_KEY=test-key` ke saath chalayein.

> ORS key sirf `backend/.env` mein likhein, `.env.example` mein nahi; woh file GitHub par jati hai.

Viva ki tayyari

Aam sawal aur chhote jawab.

Rules browser aur server dono par kyun?

Browser par foran bataane ke liye. Asal hifazat server par hai, kyunke koi bhi seedha API ko request bhej sakta hai. Dono `shared/` ke ek hi rules use karte hain.

Budget ka kharcha save kyun nahi karte?

Har dafa fuel entries jod kar nikalte hain. Entry edit ya delete ho to budget foran sahi; do jagah alag numbers ka khatra nahi.

Sab se kam fuel wala raasta kaise chuna?

Litre = km ÷ km/l, is liye sab se chhota raasta. Magar sirf un raaston mein se jo sab se tez raaste ke 1.5 guna waqt ke andar hon.

Mileage kahan se aati hai?

Odometer readings se (full-tank method). 12 mahine mein 300 km se kam naape gaye hon to gaari ki likhi km/l.

Google Maps kyun nahi?

Us ke liye billing account chahiye. OpenStreetMap aur OpenRouteService free hain; route ki key server par rehti hai.

Motorcycle ka raasta motorway par kyun nahi?

Pakistan mein motorcycle aur rickshaw motorway par mana hain, is liye un ke raaste motorway se bachte hain.

Route service band ho to?

Trip phir bhi save hota hai, wajah likhi aati hai, aur "Try again". User sign out nahi hota.

Crisis mode kaun on karta hai, aur Accept kya karta hai?

SRS kehta hai "allow users", is liye har user apne liye. Accept trip ka status badalta hai (Postponed/Combined) ya raasta; kuch delete nahi hota, "Plan again" se wapas.

Ek hi notification baar baar kyun nahi banti?

Har waqiye ki ek key hai (jaise `budget-over:2026-10:overall`). Save upsert se hota hai aur database mein `{ user, key }` unique hai, is liye do requests ek saath aayen tab bhi ek hi notification.

"Inefficient trip" kis ko kaha?

SRS mein tareef nahi. Hamari: chuna hua raasta recommended se 15% ya zyada fuel le. Ek hi gaari mein fuel km ke saath barhta hai, is liye km ka muqabla, jo electric par bhi chalta hai. 15% ek constant (`INEFFICIENT_ROUTE_PERCENT`) mein hai.

Notification foran kaise aati hai? WebSocket hai?

Nahi. Har save ke baad `api.js` ghanti ko dobara check karne ko kehta hai, aur ghanti har minute aur window par wapas aane par bhi check karti hai. Prototype ke liye kaafi aur samajhna aasaan.

Admin kaise banta hai?

Register se nahi. `npm run seed:admin` ya pehle se maujood admin.

Token chori ho jaye to?

1 din mein expire. User delete ho to agli request par 401. Password hash hai, asal kahin nahi.

Electric gaari ka kya?

Fuel entries nahi; trips mein raasta, km aur waqt, litre aur rupay nahi; crisis mein sirf km ki bachat.

Validation kahan likhi hai?

`shared/validation.js` : har check `""` (sahi) ya error message deta hai. Backend `utils/validation.js` se, frontend `src/validation.js` se lete hain.

Kya baqi hai

CODE

- Phase 7: Postman collection, demo data, har edge case ka haath se test, supervisor ke saath demo.
- Pakistan ki sarhad ka check: abhi seedha rectangle hai, is liye Afghanistan ke kuch hisse bhi maan liye jate hain.

AAP KE KARNE KE

- Asli ORS key ke saath ek dafa: Lahore → Islamabad, car aur motorcycle (motorcycle M-2 par nahi).
- Supervisor se confirm: Admin trips sirf dekhe; crisis mode har user ka apna; "kam fuel = kam km" ka andaza theek hai; inefficient trip 15%; notifications sirf app ke andar (email nahi).
- Faisla: main trip postpone ho to us mein combine kiye trips ka kya ho.

Smart Fuel Optimization & Resource Management System · CS619 · Code ki haalat 13 Oct 2026 (Phase 0 se 5 mukammal).